Event-Driven Architecture

Software Architecture

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March 30, 2026

Definition 0. Event

Something that has happened or needs to happen.

Definition 0. Event Handling

Responding to notification of an event.

Definition 0. Asynchronous Communication

Sending a message to a receiver and not waiting for a response.

Comment on how this enables parallel processing.

Responsiveness

- Synchronous Communication
  - Send message
  - *Wait* for response
  - Continue processing

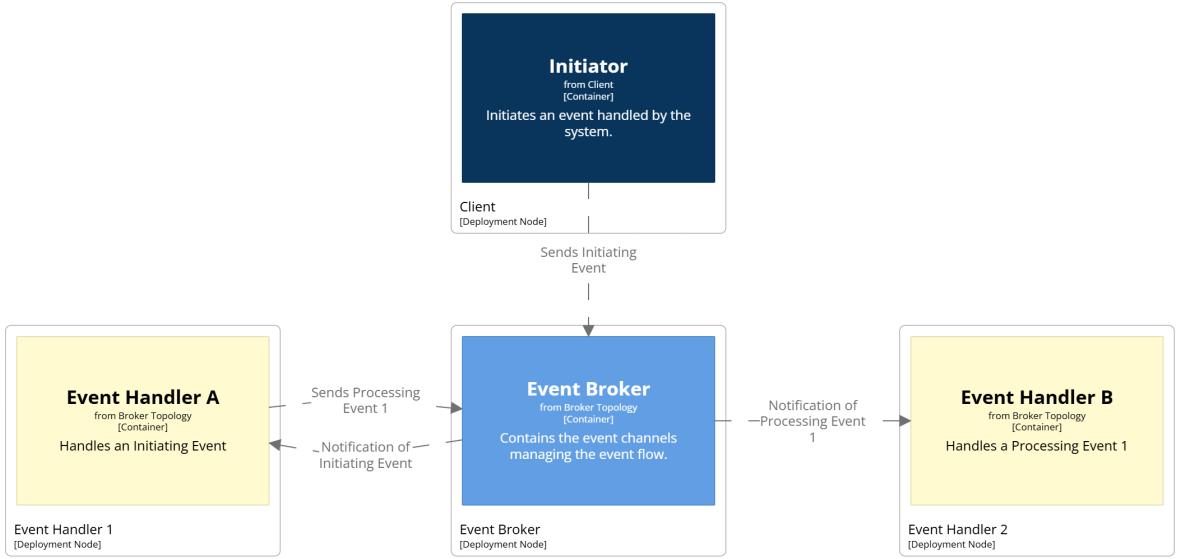
Responsiveness

- Synchronous Communication
  - Send message
  - *Wait* for response
  - Continue processing
- Asynchronous Communication
  - Send message
  - Continue processing
  - *Optionally* receive response
  - *Complex* error handling

Definition 0. Event-Driven Architecture

Asynchronous distributed system that uses event processing to *coordinate* actions in a larger business process.

Event-Driven Architecture



Comment on how each container is deployed in its own compute node.

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Terminology

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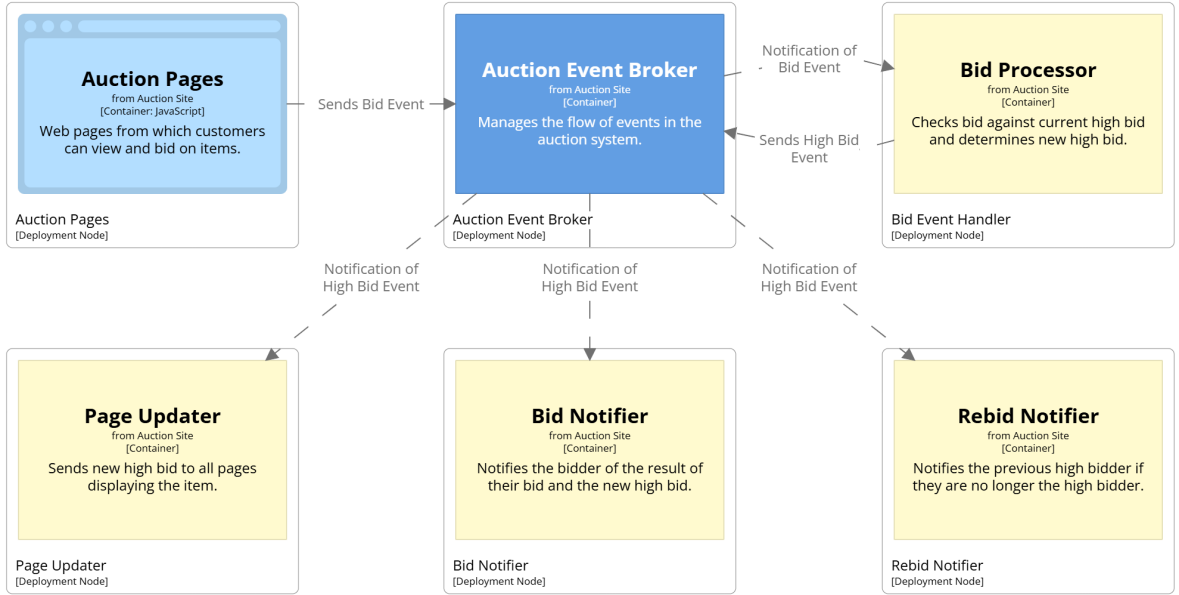
Processing Event	Indicates next step in the process can be performed
Event Channel	Holds events waiting to be processed
Event Handler	Processes an event • Step, or part of a step, in the business process

Auction Example



- Auction Event Broker has an API Gateway component to receive client requests and components to manage the event channels.
- Step through event process.
- Highlight asynchronous messages and parallel processing.
- Bid Processor could send back a high bid event or an async message.

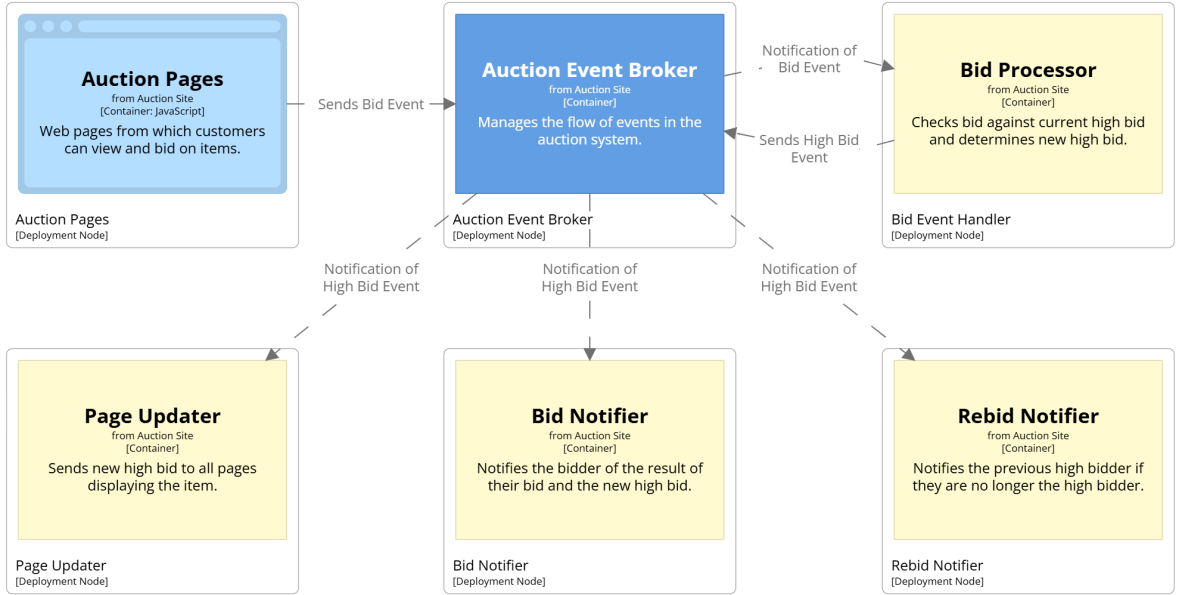
Definition 0. Event Handler Cohesion Principle

Each event handler is a simple cohesive unit that performs a *single* processing task.

Definition 0. Event Handler Independence Principle

Event handlers should not depend on the *implementation* of any other event handler.

Auction Example – Error Handling



- Ask:
- How to handle Bid Processor failing?
  - Need to restart & recover
- How to handle Rebid Notifier failing?
  - Need to restart – Could losing events be acceptable?
- How to handle Event Broker failing?
  - Need to restart & recover – without losing events

Broker All events received by event broker

- Notifies event handlers of events
- Event handlers send processing events when they finish processing

Topologies

Broker All events received by event broker

- Notifies event handlers of events
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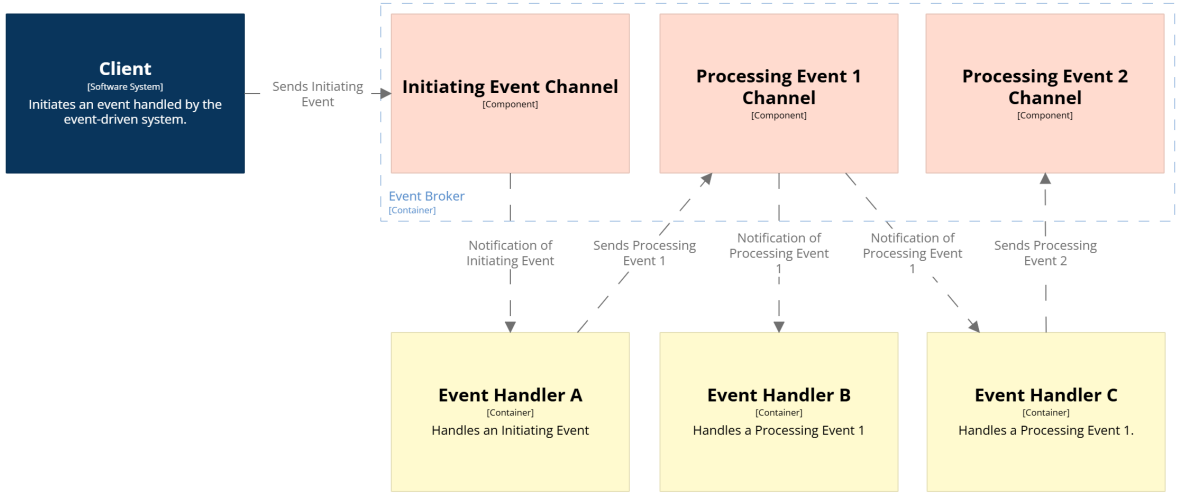
Mediator Manages business process

- Event queue of initiating events
- Event mediator sends processing events to event handlers
- Event handlers send async messages to mediator to report process finished

Broker Topology

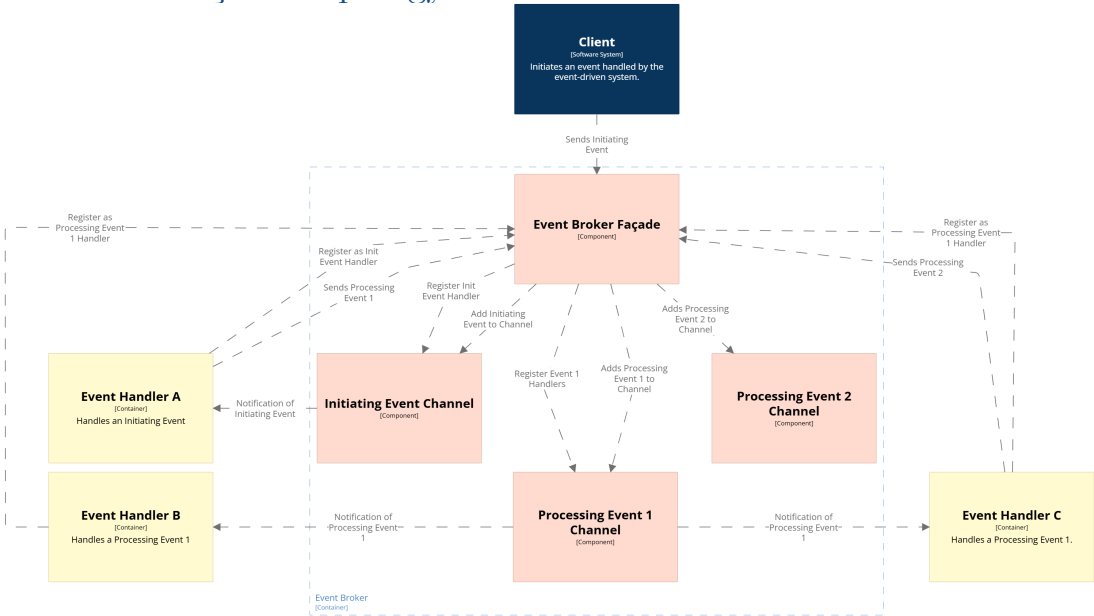


- Step through event process
- Channels facilitate message flow
  - Commonly a lightweight message broker (e.g. RabbitMQ, ...)
- Send final processing event, even if it is not handled
  - Easier to *extend* in the future

Event Broker Façade

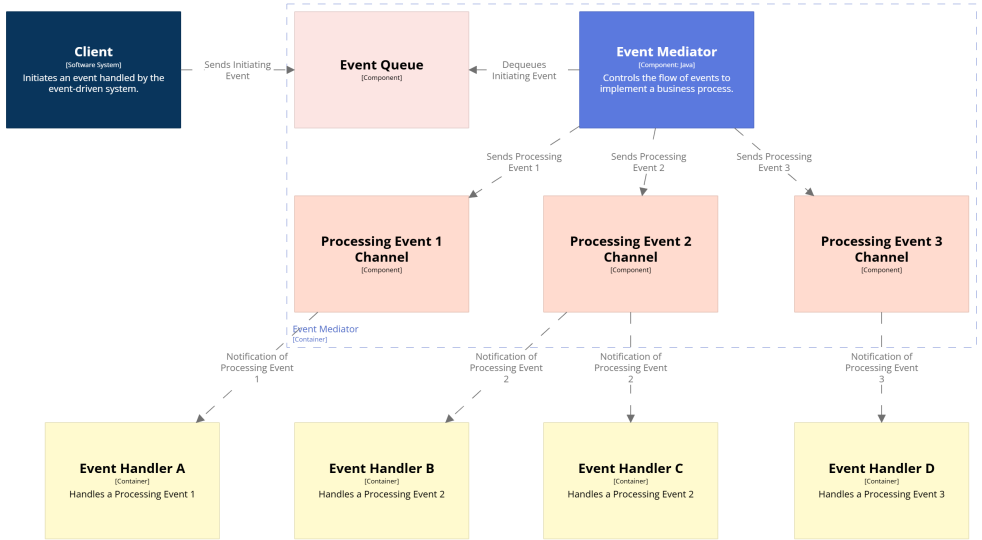
- Event handlers register to *listen* for events
- Receives events and *directs* them to the correct channel

Broker with Façade Topology



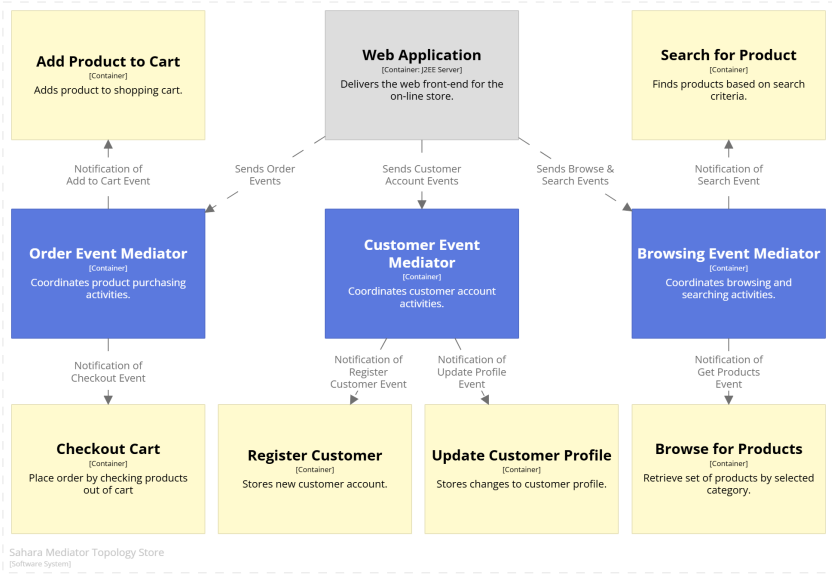
- Event processing & event handling are the same
- Event Handlers register to listen for events, rather than being connected directly to Channels
  - Additional layer of abstraction
- Step through event process

Mediator Topology



- Step through event process.
- Highlight process control performed by mediator.

Sahara Mediator Topology



- Step through event processes.
  - Note that external clients are not shown in diagram
- Multiple mediators is common – *one* per domain.
- Discuss internals of mediators.
  - Event queue and event channels

Extensibility

- New behaviour for existing event
  - Broker: Implement event handler & register with broker
    - Existing ignored event hooks
  - Mediator: Implement event handler & modify mediator logic

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- New behaviour for existing event
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- New event
  - Broker: Implement event & event handler, create event channel, modify broker façade
  - Mediator: Implement event & event handler, modify mediator logic

Scalability

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Scalability

- Event handlers deployed independently
  - Scaled independently to manage load
- Event broker federated
  - Distributed across multiple compute nodes
- Event mediators for different domains
  - Distributes loads by domain
    (e.g. browse & search, account, & order events)
    - Scaled independently to manage load

Queues

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  - FIFO behaviour

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- Multiple front of queue pointers
  - For each event handler
- Event removed when event handlers finish
  - Retry if a handler fails
- Events persist until removed
  - Recovery from broker failure

Streams

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- Handlers notified when event added to stream
  - Observer pattern
- Handlers process events at their own pace
  - Cardiac arrest alarm vs. heart rate graph
- Events history
  - Redo processing
  - Review processing activities

Queues vs. Streams

- Queue
  - Known steps in business process
  - Easier sequencing of steps in business process
  - “Exactly once” semantics
  - eCommerce system

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- Queue
  - Known steps in business process
  - Easier sequencing of steps in business process
  - “Exactly once” semantics
  - eCommerce system
- Stream
  - Very large number of events or handlers
  - Handlers can ignore events
  - Analysis of past activity
  - Event sourcing

Broker dumb pipe

Broker events have occurred

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- Broker events have occurred

- Mediator smart pipe
- Mediator events are commands to process

Broker Advantages

- Scalability
- Reliability
- Extensibility
- Low coupling

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Mediator Advantages

- Complex business process logic
- Error handling
- Maintain process state
- Error recovery

Emphasise that the *real* advantage of Broker is *low coupling* and slightly easier *extensibility*.

Pros & Cons

- Modularity Event Handlers
- Extensibility
- Reliability Event Handlers
- Interoperability Events
- Scalability Event Handlers
- Security
- Simplicity
- Deployability
- Testability Complex Interactions

- Broker & Mediator are both *very* scalable
- Due to simple event management through message queues
  - Broker can handle a *slightly higher load*
  - Broker is *slightly easier to scale*
- Mediator has more *internal processing*, so requires greater *resources*